Topic 11: Kinetics

Knowledge of the concepts introduced in Unit 2, Topic 9A: Kinetics will be assumed and extended in this topic.

Students will be assessed on their ability to:

11.1	understand the terms: i rate of reaction ii rate equation, rate=k[A]^m[B]^n where m and n are 0, 1 or 2 iii order with respect to a substance in a rate equation iv overall order of a reaction v rate constant vi half-life vii rate-determining step viii activation energy ix heterogeneous and homogeneous catalyst
11.2	be able to calculate the half-life of a reaction, using data from a suitable graph, and identify a reaction with a constant half-life as being first order
11.3	be able to select and justify a suitable experimental technique to obtain rate data for a given reaction, including: i titration ii colorimetry iii mass change iv volume of gas evolved v other suitable technique(s) for a given reaction
11.4	understand experiments that can be used to investigate reaction rates by: i an initial-rate method, carrying out separate experiments where different initial concentrations of one reagent are used *A 'clock reaction' is an acceptable approximation of this method.* ii a continuous monitoring method to generate data to enable concentration-time or volume-time graphs to be plotted
11.5	be able to deduce the order (0, 1 or 2) with respect to a substance in a rate equation, using data from: i a concentration-time graph ii a rate-concentration graph iii an initial-rate method

11.6	understand how to: i obtain data to calculate the order with respect to the reactants (and the hydrogen ion) in the acid-catalysed iodination of propanone ii use these data to make predictions about species involved in the rate-determining step iii deduce a possible mechanism for the reaction
11.7	be able to deduce the rate-determining step from a rate equation and vice versa
11.8	be able to deduce a reaction mechanism, using knowledge of the rate equation and the stoichiometric equation for a reaction
11.9	understand that knowledge of the rate equations for the hydrolysis of halogenoalkanes can be used to provide evidence for S_N1 and S_N2 mechanisms for tertiary and primary halogenoalkane hydrolysis
11.10	be able to use calculations and graphical methods to find the activation energy for a reaction from experimental data *The Arrhenius equation will be given if needed.*
11.11	understand the use of a solid (heterogeneous) catalyst for industrial reactions, in the gas phase, in terms of providing a surface for the reaction
11.12	**CORE PRACTICALS 9a and 9b** **Following the rate of the iodine-propanone reaction by a titrimetric method and investigating a 'clock reaction' (Harcourt-Esson, iodine clock).**
11.13	**CORE PRACTICAL 10** **Finding the activation energy of a reaction.**
	Further suggested practicals: i the reaction between marble chips and hydrochloric acid (change of mass or change in volume of gas) ii the reaction between magnesium and hydrochloric acid to determine the activation energy iii following the rate of the iodine-propanone reaction by a colorimetric method iv the catalysis by a cobalt(II) salt of potassium sodium tartrate and hydrogen peroxide v the action of the enzyme urease on urea and thiourea